Chapter 2: Series of Functions

Calculus II

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Section 1

Power Series

Power Series

P-Series

Problem: For which values of x does the series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converge?

Solution: Observe that if $b_n = \left|\frac{x^n}{n}\right|$ then

$$\begin{aligned}\lim_{n\to\infty} \frac{b_{n+1}}{b_n} &= \lim_{n\to\infty} \frac{\frac{|x|^{n+1}}{n+1}}{\frac{|x|^n}{n}} \\ &= \lim_{n\to\infty} |x| \frac{n}{n+1} \\ &= |x|\end{aligned}$$

$\begin{cases} x \in [-1, 1) \\ \text{Converges} \\ \\ \text{Diverges - Otherwise} \end{cases}$

The Ratio Test shows that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ converges absolutely if $|x| < 1$ and the series diverges if $|x| > 1$.

If $x = 1$, the series becomes $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges.

If $x = -1$, the series becomes $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$, which converges.

Power Series

Definition 2.1.1 (Power Series)

A power series centered at $x = a$ is a formal series of the form

$$\sum_{n=0}^{\infty} a_n (x-a)^n$$

where x is viewed as a variable.
The value a_n is called the coefficient of the term $(x-a)^n$.

Central Questions:

1. If $I = \{x_0 \in \mathbb{R} \mid \sum_{n=0}^{\infty} a_n (x_0 - a)^n \text{ converges}\}$, then what can we say about I?
2. If we define a function f on I by $f(x_0) = \sum_{n=0}^{\infty} a_n (x_0 - a)^n$ for all $x_0 \in I$, then what can we say about $f(x)$?

Power Series

Observations:

1. Our convention is that $0^0 = 1$, so that if $x = a$, the series becomes

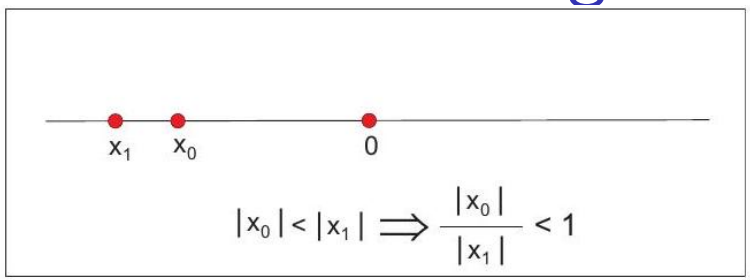
$$\sum_{n=0}^{\infty} a_n(a-a)^n = a_0 + 0 + 0 + 0 + \cdots = a_0$$

1. We can let $u = x - a$ to get that $\sum_{n=0}^{\infty} a_n(x-a)^n$ converges at $x = x_0$ if and only if $\sum_{n=0}^{\infty} a_n u^n$ converges at $u = x_0 - a$.

Consequence: We need only focus on series of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

Interval of Convergence



Note: The series $\sum_{n=0}^{\infty} a_n x^n$ always converges at $x = 0$.
Assume that $\sum_{n=0}^{\infty} a_n x_1^n$ converges where $x_1 \neq 0$.

Then $\lim_{n\to\infty} |a_n x_1^n| = 0$ and there exists an $N_0 \in \mathbb{N}$ so that if $n \geq N_0$ we have $|a_n x_1^n| \leq 1$.
Let $|x_0| < |x_1|$. Then if $n \geq N_0$

$$|a_n x_0^n| = |a_n x_1^n| \left|\frac{x_0}{x_1}\right|^n \leq \left|\frac{x_0}{x_1}\right|^n$$

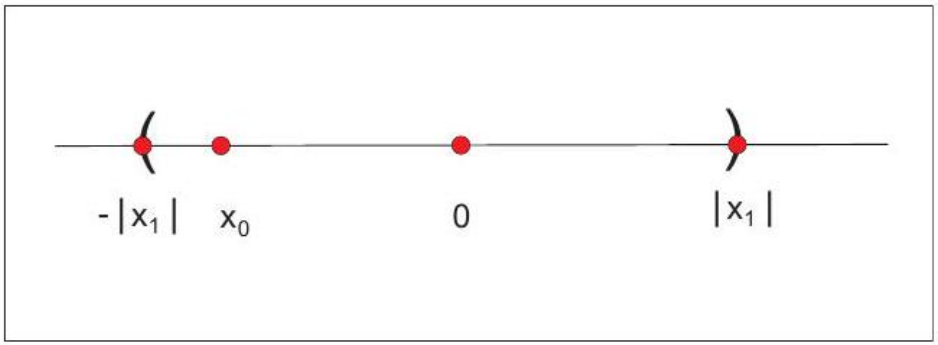
Since $\left|\frac{x_0}{x_1}\right| < 1$, the series

$$\sum_{n=N_0}^{\infty} \left|\frac{x_0}{x_1}\right|^n$$

converges. By the Comparison Test we have that

$$\sum_{n=0}^{\infty} a_n x_0^n \quad \text{converges absolutely.}$$

Interval of Convergence



Key Observations: If the power series $\sum_{n=0}^{\infty} a_n x^n$ converges at $x_1 \neq 0$ and if $|x_0| < |x_1|$, then the series converges at x_0 as well. Let

$$I = \left\{ x_0 \in \mathbb{R} \mid \sum_{n=0}^{\infty} a_n x_0^n \text{ converge} \right\}$$

Then

$$(-|x_1|, |x_1|) \subset I$$

$\Rightarrow I$ is an interval centered around $x = 0$.

Note: The same analysis works for power series of the form $\sum_{n=0}^{\infty} a_n (x-a)^n$.

Interval and Radius of Convergence

Definition 2.1.2 (Interval and Radius of Convergence)

Given a power series of the form $\sum_{n=0}^{\infty} a_n(x-a)^n$, the set

$$I = \left\{ x_0 \in \mathbb{R} \mid \sum_{n=0}^{\infty} a_n \left(x_0 - a\right)^n \text{ converges} \right\}$$

is an interval centered at $x = a$ which we call the interval of convergence for the power series. Let

$$R = \begin{cases} \operatorname{lub}\left(\{|x_0 - a| \mid x_0 \in I\}\right) & \text{if } I \text{ is bounded} \\ \infty & \text{if } I \text{ is not bounded.} \end{cases}$$

Then R is called the radius of convergence of the power series.

Interval of Convergence

Theorem 2.1.3 (Fundamental Convergence Theorem for Power Series)

Given a power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ centered at $x=a$, let R be the radius of convergence.

1. *If $R=0$, then $\sum_{n=0}^{\infty} a_n(x-a)^n$ converges for $x=a$, but it diverges for all other values of x.*
2. *If $0<R<\infty$, then the series $\sum_{n=0}^{\infty} a_n(x-a)^n$ converges absolutely for every $x \in (a-R, a+R)$ and diverges if $|x-a|>R$.*
3. *If $R=\infty$, then the series $\sum_{n=0}^{\infty} a_n(x-a)^n$ converges absolutely for every $x \in \mathbb{R}$.*

Interval of Convergence

Remark: If $0 < R < \infty$, then there are four possibilities for the interval of convergence I.

a - centre

$x \neq -1$ or 1

1) $I = (a - R, a + R)$ Example: $\sum_{n=0}^{\infty} x^n \Rightarrow I = (-1, 1)$.

2) $I = [a - R, a + R)$ Example: $\sum_{n=1}^{\infty} \frac{x^n}{n} \Rightarrow I = [-1, 1)$.

3) $I = (a - R, a + R]$ Example: $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n} \Rightarrow I = (-1, 1]$.

4) $I = [a - R, a + R]$ Example: $\sum_{n=1}^{\infty} \frac{x^n}{n^2} \Rightarrow I = [-1, 1]$.

Key Note: Once R is determined, you need to test the endpoints separately.

Radius of Convergence

Question: How do we find the radius of convergence R?
Key Observation: Given $\sum_{n=0}^{\infty} a_n x^n$, assume that

$$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

where $0 \leq L < \infty$. For $x_0 \neq 0$, let

$$b_n = a_n x_0^n$$

then

$$\begin{aligned}
\lim_{n\to\infty} \left| \frac{b_{n+1}}{b_n} \right| &= \lim_{n\to\infty} \left| \frac{a_{n+1} x_0^{n+1}}{a_n x_0^n} \right| \\
&= \lim_{n\to\infty} |x_0| \left| \frac{a_{n+1}}{a_n} \right| \\
&= |x_0| \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \\
&= L\,|x_0|
\end{aligned}$$

Radius of Convergence

Conclusions:

1. The Ratio Test shows that the series $\sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} a_n x_0^n$ converges absolutely if $L\,|x_0| < 1$ and diverges if $L\,|x_0| > 1$.
2. Assume that $0 < L < \infty$. Then $L\,|x_0| < 1$ if and only if $|x_0| < \frac{1}{L}$. Therefore, the radius of convergence is $\frac{1}{L}$.
3. If $L = 0$, then no matter the value of x_0, we have $L\,|x_0| = 0 < 1$. Therefore, $R = \infty$.
4. If

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$$

then the same calculation would show that

$$\lim_{n\to\infty} \left| \frac{b_{n+1}}{b_n} \right| = \infty$$

and as such the series diverges for all nonzero x_0. But we know that the series must converge at $x = 0$, so $R = 0$.

Radius of Convergence

Theorem 2.1.4 (Test for the Radius of Convergence)

Let $\sum_{n=0}^{\infty} a_n(x-a)^n$ *be a power series for which*

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L$$

where $0 \leq L < \infty$ *or* $L = \infty$. *Let* R *be the radius of convergence of the power series.*

1. *If* $0 < L < \infty$, *then* $R = \frac{1}{L}$.
2. *If* $L = 0$, *then* $R = \infty$.
3. *If* $L = \infty$, *then* $R = 0$.

Example 2.1.5

Find the radius and interval of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{x^n}{3^n \left(n^2+1\right)}$$

Cont'd

Solution: Let $a_n = \frac{1}{3^n(n^2+1)}$. Then

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{\frac{1}{3^{n+1}((n+1)^2+1)}}{\frac{1}{3^n(n^2+1)}}\right| \\
&= \lim_{n\to\infty}\left|\frac{3^n\left(n^2+1\right)}{3^{n+1}\left((n+1)^2+1\right)}\right| \\
&= \lim_{n\to\infty}\frac{1}{3}\left(\frac{n^2+1}{n^2+2n+2}\right) \\
&= \frac{1}{3}\lim_{n\to\infty}\frac{1+\frac{1}{n^2}}{1+\frac{2}{n}+\frac{2}{n^2}} \\
&= \frac{1}{3}
\end{aligned}$$

It follows from the Radius of Convergence Test that $R = \frac{1}{\frac{1}{3}} = 3$.

Cont'd

For $x = 3$, the series becomes

$$\sum_{n=0}^{\infty} \frac{3^n}{3^n\left(n^2+1\right)} = \sum_{n=0}^{\infty} \frac{1}{n^2+1}$$

Since

$$\frac{1}{n^2+1} < \frac{1}{n^2}$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the Comparison Test shows that $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges and hence that

$$\sum_{n=0}^{\infty} \frac{1}{n^2+1}$$

converges.

Cont'd

Similarly, if $x = -3$, the series becomes

$$\sum_{n=0}^{\infty} \frac{(-3)^n}{3^n\left(n^2+1\right)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$$

Then the Alternating Series Test applies and this series also converges. Alternately, we have

$$\left|\frac{(-1)^n}{n^2+1}\right| = \frac{1}{n^2+1}$$

and since $\sum_{n=0}^{\infty} \frac{1}{n^2+1}$ converges, this shows $\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$ converges absolutely.

Therefore, the interval of convergence is $[-3, 3]$.

Example 2.1.6

Find the radius and interval of convergence for the series

$$1 + 2x + x^2 + 2x^3 + x^4 + \cdots$$

In this case

$$\frac{a_{n+1}}{a_n} = \begin{cases} 2 & \text{if } n \text{ is even} \\ \frac{1}{2} & \text{if } n \text{ is odd} \end{cases}$$

so $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|}$ does not exist.

Observations:

1. The Divergence Test shows the series diverges if $|x| \geq 1$.
2. If $0 \leq x_0 < 1$, then

$$0 < 1 + 2x_0 + x_0^2 + 2x_0^3 + x_0^4 + \cdots \leq 2\left(1 + x_0 + x_0^2 + x_0^3 + x_0^4 + \cdots\right)$$

   so the series converges by Comparison Test and Geometric Series Test.

Hence $R = 1$ and the interval of convergence is $(-1, 1)$.

Functions Represented by Power Series

Example 2.1.7

Recall that the geometric series

$$\sum_{n=0}^{\infty} x^n$$

converges for each x with $|x| < 1$.
Moreover, if $|x| < 1$, then we also know that

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Note: This means that the series provides us with a means to represent the function $f(x) = \frac{1}{1-x}$ on the interval $(-1, 1)$.

Functions Represented by Power Series

Definition 2.1.8 (Functions Represented by a Power Series)

Let $\sum_{n=0}^{\infty} a_n(x-a)^n$ be a power series with radius of convergence $R > 0$. Let I be the interval of convergence for $\sum_{n=0}^{\infty} a_n(x-a)^n$.
Let f be the function defined on the interval I by the formula

$$f(x_0) = \sum_{n=0}^{\infty} a_n (x_0 - a)^n \qquad (*)$$

for each $x_0 \in I$.
We say that the function f is represented by the power series

$$\sum_{n=0}^{\infty} a_n(x-a)^n \text{ on } I.$$

Example 2.1.9

Find a power series representation for the function

$$f(x) = \frac{x}{1-x^2}$$

on the interval $(-1, 1)$.

Solution: We know for any $-1 < u < 1$ that

$$\frac{1}{1-u} = \sum_{n=0}^{\infty} u^n$$

If $-1 < x < 1$, and if $u = x^2$, then $u \in (-1, 1)$, so

$$\frac{1}{1-x^2} = \sum_{n=0}^{\infty} \left(x^2\right)^n = \sum_{n=0}^{\infty} x^{2n}$$

Hence

$$\frac{x}{1-x^2} = x \cdot \sum_{n=0}^{\infty} x^{2n} \quad = x \cdot \lim_{k\to\infty} \sum_{n=0}^{k} x^{2n} = \lim_{k\to\infty} \sum_{n=0}^{k} x^{2n+1} = \sum_{n=0}^{\infty} x^{2n+1}$$

Example 2.1.10

Find the function represented by the power series

$$1 + 2x + x^2 + 2x^3 + x^4 + 2x^5 + \cdots$$

on the interval $(-1, 1)$.

Observation: If $x \in (-1, 1)$, then

$$\begin{aligned}
1 + 2x + x^2 + 2x^3 + x^4 + \cdots &= \left(1 + x + x^2 + x^3 + \cdots\right) \\
&\quad + \left(x + x^3 + x^5 + \cdots\right) \\
&= \sum_{n=0}^{\infty} x^n + \sum_{n=0}^{\infty} x^{2n+1} \\
&= \frac{1}{1-x} + \frac{x}{1-x^2} \\
&= \frac{1+x}{1-x^2} + \frac{x}{1-x^2} \\
&= \frac{1+2x}{1-x^2}
\end{aligned}$$

for all $x \in (-1, 1)$.

Functions Represented by Power Series

Question: Assume that the series $\sum_{n=0}^{\infty} a_n(x-a)^n$ has radius of convergence $R > 0$ and interval of convergence I. What are the properties of the function

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n?$$

1. Is f continuous on I?
2. Is f differentiable on $(-R, R)$? And if so what is its derivative?

Functions Represented by Power Series

Theorem 2.1.11 (Abel's Theorem: Continuity of Power Series)

Assume that the power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ *has interval of convergence I. Let*

$$f(x_0) = \sum_{n=0}^{\infty} a_n (x_0 - a)^n$$

for each $x_0 \in I$. *Then* f *is continuous on* I.

Functions Represented by Power Series

Question: Suppose that f and g are represented by power series centered at $x = a$ of the form

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n \text{ and } g(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$$

with intervals of convergence I_f and I_g respectively.
Can this information be used to build a power series representation for $f + g$?
Observation: If $x_0 \in I_f \cap I_g$, then

$$\begin{aligned}
(f+g)(x_0) &= f(x_0) + g(x_0) \\
&= \lim_{k\to\infty} \sum_{n=0}^{k} a_n (x_0 - a)^n + \lim_{k\to\infty} \sum_{n=0}^{k} b_n (x_0 - a)^n \\
&= \lim_{k\to\infty} \sum_{n=0}^{k} (a_n + b_n)(x_0 - a)^n \\
&= \sum_{n=0}^{\infty} (a_n + b_n)(x_0 - a)^n
\end{aligned}$$

Theorem 2.1.12 (Addition of Power Series)

Assume that f and g are represented by power series centered at $x = a$ with

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n \quad \text{and} \quad g(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$$

respectively.
Assume also that the radii of convergence of these series are R_f and R_g with intervals of convergence I_f and I_g. Then if $x \in I_f \cap I_g$,

$$(f+g)(x) = \sum_{n=0}^{\infty} (a_n + b_n)(x-a)^n$$

Moreover:

- *if $R_f \neq R_g$, then the radius of convergence of the power series representing $f+g$ is $R = \min\{R_f, R_g\}$ and the interval of convergence is $I = I_f \cap I_g$.*
- *If $R_f = R_g$, then $R \geq R_f$.*

Functions Represented by Power Series

Question: Assume that the series $\sum_{n=0}^{\infty} a_n(x-a)^n$ has radius of convergence $R > 0$ and interval of convergence I. What are the properties of the function

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n?$$

1. Is f continuous on I? Yes, this follows from Abel's Theorem.
2. Is f differentiable on $(a-R, a+R)$?

Differentiation of Power Series

Strategy: If we have a function

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

that is represented by a power series with radius of convergence $R > 0$, we could try to differentiate f by differentiating the series one term at a time.
Since

$$\frac{d}{dx}\left(a_n (x-a)^n\right) = n a_n (x-a)^{n-1},$$

we get:

Definition 2.1.13 (Formal Derivative of a Power Series)

Given a power series $\sum_{n=0}^{\infty} a_n (x-a)^n$, the formal derivative is the series

$$\sum_{n=1}^{\infty} n a_n (x-a)^{n-1}.$$

Differentiation of Power Series

Two Fundamental Problems:

Problem 1: For which values of x does the formal power series

$$\sum_{n=1}^{\infty} n a_n (x-a)^{n-1}$$

converge? In particular, does this series converge for the same values as the original series $\sum_{n=0}^{\infty} a_n (x-a)^n$?

Problem 2: If both of the series $\sum_{n=0}^{\infty} a_n (x-a)^n$ and $\sum_{n=1}^{\infty} n a_n (x-a)^{n-1}$ converge at the same x, must it be the case that

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-a)^{n-1}?$$

Differentiation of Power Series

Problem 1: For which values of x does the formal power series

$$\sum_{n=1}^{\infty} n a_n (x-a)^{n-1}$$

converge? In particular, does this series converge for the same values as the original series $\sum_{n=0}^{\infty} a_n (x-a)^n$?

Observation: The series $\sum_{n=0}^{\infty} a_n (x-a)^n$ and the series $\sum_{n=0}^{\infty} n a_n (x-a)^n$ have the same radius of convergence.
We can show that the series $\sum_{n=0}^{\infty} a_n (x-a)^n$ and its formal derivative $\sum_{n=1}^{\infty} n a_n (x-a)^{n-1}$ also have the same radius of convergence, though the interval of convergence may be different. Therefore,

$$g(x) = \sum_{n=1}^{\infty} n a_n (x-a)^{n-1}$$

is defined for all $x \in (a-R, a+R)$. Is $g(x) = f'(x)$?

Differentiation of Power Series

Theorem 2.1.14 (Differentiation of Power Series)

Assume that the power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ *has radius of convergence* $R > 0$. *Let*

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

for all $x \in (a-R, a+R)$. *Then* f *is differentiable on* $(a-R, a+R)$ *and for each* $x \in (a-R, a+R)$,

$$f'(x) = \sum_{n=1}^{\infty} n a_n(x-a)^{n-1}$$

Differentiation of Power Series

Example 2.1.15

If $|x| < 1$, then let

$$f(x) = \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Differentiating term-by-term, we get

$$f'(x) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1}$$

Question: Evaluate

$$\sum_{n=1}^{\infty} \frac{n}{2^{n-1}}$$

Observation: This series is obtained from $\sum_{n=1}^{\infty} nx^{n-1}$ by letting $x = \frac{1}{2}$. Therefore,

$$\sum_{n=1}^{\infty} \frac{n}{2^{n-1}} = f'\left(\frac{1}{2}\right) = \frac{1}{\left(1-\frac{1}{2}\right)^2} = 4$$

Example 2.1.16

For any $x \in \mathbb{R}$ let

$$g(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Term-by-term differentiation gives

$$g'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!}$$
$$= \sum_{n=0}^{\infty} \frac{x^n}{n!} = g(x)$$

Hence

$$g(x) = Ce^x$$

for some constant C. However, $C = g(0) = 1$, so

$$g(x) = e^x$$

Example 2.1.17

Find a power series representation for the function $f(x) = e^{-x^2}$.
We have that for any $u \in \mathbb{R}$ that

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!} \qquad (*)$$

Let $u = -x^2$ and plug this in (*) to get for all $x \in \mathbb{R}$

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}$$

Note: It may look like

$$e^{-x^2} = 1 - \frac{x^2}{1!} + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots + (-1)^n \frac{x^{2n}}{n!} + \cdots$$

is not a power series since there are no terms involving x^n when n is odd. But in fact, it really is a power series where the coefficients are of the form $a_{2k-1} = 0$ and $a_{2k} = (-1)^k \frac{1}{(k)!}$ for each $k = 0, 1, 2, 3, 4, \ldots$

Uniqueness of Power Series Representations

Question: Can a function f have more than one power series representation centered at $x = a$ with $R > 0$? That is, if

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

and

$$f(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$$

must

$$a_n = b_n$$

for all $n \in \mathbb{N} \cup \{0\}$?

Uniqueness of Power Series Representations

Observation: If $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n = \sum_{n=0}^{\infty} b_n(x-a)^n$, we have

$$a_0 = f(a) = b_0$$

Since f is differentiable on $(a-R, a+R)$ with

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-a)^{n-1} = a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + \cdots$$

and

$$f'(x) = \sum_{n=1}^{\infty} n b_n (x-a)^{n-1} = b_1 + 2b_2(x-a) + 3b_3(x-a)^2 + \cdots$$

we get that

$$a_1 = f'(a) = b_1$$

Important Note: f' is also differentiable on $(a-R, a+R)$.

Uniqueness of Power Series Representations

Observation (continued): Since

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)a_n(x-a)^{n-2} = 2a_2 + 3 \cdot 2a_3(x-a) + \cdots$$

and

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)b_n(x-a)^{n-2} = 2b_2 + 3 \cdot 2b_3(x-a) + \cdots$$

we get that

$$2a_2 = f''(a) = 2b_2$$

Hence

$$a_2 = \frac{f''(a)}{2} = b_2$$

Uniqueness of Power Series Representations

Observation (continued): In fact f has derivatives of all orders on $(a-R, a+R)$ with

$$f'''(x) = \sum_{n=3}^{\infty} n(n-1)(n-2)a_n(x-a)^{n-3}$$

and

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)(n-3)\cdots(n-k+1)a_n(x-a)^{n-k}.$$

Hence

$$f'''(a) = 3!a_3 \Rightarrow a_3 = \frac{f'''(a)}{3!}$$

and

$$f^{(k)}(a) = k!a_k \Rightarrow a_k = \frac{f^{(k)}(a)}{k!}$$

Note: We would also have

$$b_k = \frac{f^{(k)}(a)}{k!}.$$

Theorem 2.1.18 (Uniqueness of Power Series Representations)

Suppose that

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

for all $x \in (a-R, a+R)$ *where* $R > 0$. *Then*

$$a_n = \frac{f^{(n)}(a)}{n!}, \quad n = 0, 1, 2, \ldots$$

In particular, if we also had that

$$f(x) = \sum_{n=0}^{\infty} b_n (x-a)^n$$

then we must have that

$$b_n = a_n$$

for each $n = 0, 1, 2, \ldots$

Remark: f may have different representations at different center points.

Definition 2.1.19 (Formal Antiderivative of a Power Series)

Given a power series $\sum_{n=0}^{\infty} a_n(x-a)^n$, we define the formal antiderivative to be the power series

$$\sum_{n=0}^{\infty} \int a_n(x-a)^n dx = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

where C is an arbitrary constant.

Formal Antiderivative of a Power Series

Fundamental Problem: Suppose that the power series

$$\sum_{n=0}^{\infty} a_n(x-a)^n$$

has radius of convergence $R > 0$. Let

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

be the function that is represented by this power series on the interval $(a-R, a+R)$. Are the formal anitderivatives

$$C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

true antiderivatives of the function f?

Formal Antiderivative of a Power Series

Key Observations:

1. The series

$$\sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^n$$

is obtained from $\sum_{n=0}^{\infty} a_n(x-a)^n$ by dividing each coefficient a_n by $q(n) = n+1$ so they have the same radius of convergence R.

2. The formal antiderivative

$$C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

will have radius of convergence R.

3. If

$$F(x) = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

on $(a-R, a+R)$, then its formal derivative is

$$\sum_{n=0}^{\infty} a_n(x-a)^n$$

Theorem 2.1.20 (Term-by-Term Integration of a Power Series)

Assume that the power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ *has radius of convergence* $R > 0$. *Let*

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

for every $x \in (a-R, a+R)$. *Then the series*

$$\sum_{n=0}^{\infty} \int a_n(x-a)^n dx = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

also has radius of convergence R *and if*

$$F(x) = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

then $F'(x) = f(x)$.
Furthermore, if $[c,b] \subset (a-R, a+R)$, *then*

$$\int_c^b f(x)dx = \int_c^b \sum_{n=0}^{\infty} a_n(x-a)^n dx = \sum_{n=0}^{\infty} \int_c^b a_n(x-a)^n dx$$

Term-by-Term Integration of a Power Series

Important Note: It may seem perfectly natural that we are also able to integrate term-by-term the functions that are represented by a power series.
In general, if

$$F(x) = \sum_{n=1}^{\infty} f_n(x)$$

for each $x \in [a, b]$, then we might hope that

$$\int_a^b F(x)dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)dx$$

Fact: If we do not make any additional assumptions about the nature of the functions f_n or about how the series converges, then it is possible that the function f need not even be integrable on $[a, b]$ even if all of the f_n's are.

Example 2.1.21

Find a power series representation for $\ln(1+x)$.

Solution: Note that

$$\frac{d}{dx}(\ln(1+x)) = \frac{1}{1+x}$$

Key Observation: For any $u \in (-1,1)$,

$$\frac{1}{1-u} = \sum_{n=0}^{\infty} u^n$$

If $u = -x$ we get

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-x)^n$$

for any $x \in (-1,1)$.

Term-by-Term Integration of a Power Series

Solution (continued): Therefore, there is a constant C such that

$$\ln(1+x) = C + \sum_{n=0}^{\infty} \int (-1)^n x^n dx$$

$$= C + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

for all $x \in (-1, 1)$.
When $x = 0$ we get

$$0 = \ln(1+0)$$

$$= C + \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} 0^{n+1}$$

$$= C$$

Therefore

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

for all $x \in (-1, 1)$.

Term-by-Term Integration of a Power Series

Key Note: The series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

has radius of convergence $R = 1$. However, if $x = 1$ the series becomes

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} 1^{n+1} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$$

which is exactly the Alternating Series. Therefore, the series also converges at $x = 1$.

By Abel's Theorem, the equation

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$$

is actually valid on (-1,1] and

$$\ln(2) = \ln(1+1) = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

Section 2

Taylor Polynomials and Taylor's Theorem

Linear Approximation

Recall:
If $f(x)$ is differentiable at $x = a$, then

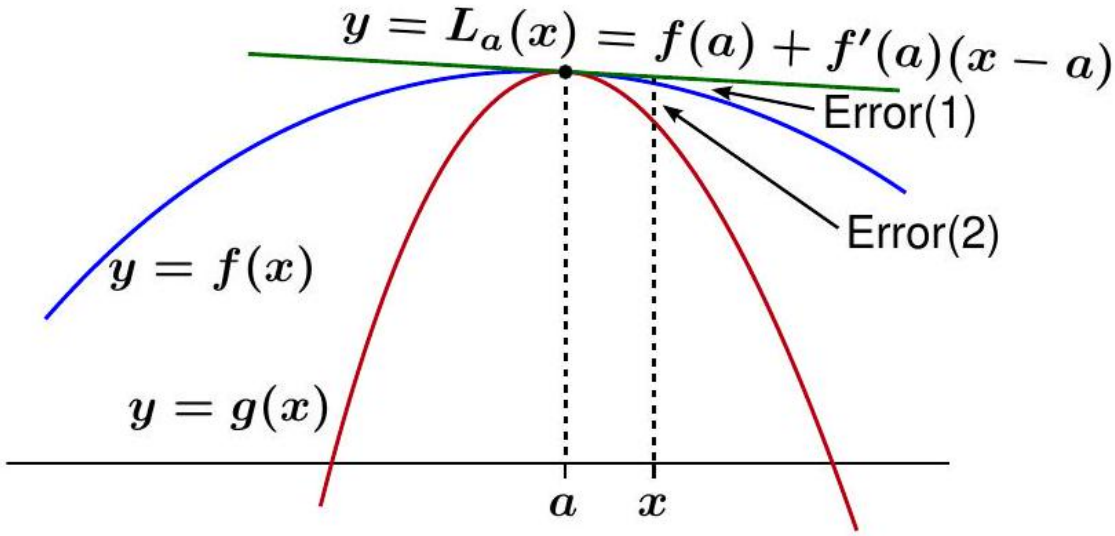
$$L_a^f(x) = f(a) + f'(a)(x - a)$$

is called the **Linear Approximation** to $f(x)$ centered at $x = a$.

Key Properties:

1. $L_a^f(a) = f(a)$.
2. $\left(L_a^f\right)'(a) = f'(a)$.
3. $L_a^f(x)$ is the unique function of the form $y = c_o + c_1(x - a)$ satisfying (1) and (2).
4. If $x \cong a$, then $L_a^f(x) \cong f(x)$.

Note: The graph of $L_a^f(x)$ is the tangent line to the graph of $f(x)$ through $(a, f(a))$.



Observation: The error in linear approximation

$$\left|f(x) - L_a^f(x)\right|$$

depends on two factors:

1. The distance from x to $a \Rightarrow |x - a|$.
2. The curvature of the graph of $f(x)$ near $x = a \Rightarrow |f''(x)|$ near $x = a$.

Question 1: Can we approximate $f(x)$ better by using a second degree polynomial that also encodes $f''(a)$?

Taylor Polynomials

Question 2: Can we find a polynomial

$$p(x) = c_0 + c_1(x-a) + c_2(x-a)^2$$

such that

$$p(a) = f(a),\ p'(a) = f'(a),\ p''(a) = f''(a)?$$

Solution:

i) $f(a) = p(a) = c_0 + c_1(a-a) + c_2(a-a)^2 = c_0 \Rightarrow c_0 = f(a).$

ii) $p'(x) = c_1 + 2 \cdot c_2(x-a) \Rightarrow f'(a) = p'(a) =$

$$c_1 + 2 \cdot c_2(a-a) = c_1 \Rightarrow c_1 = f'(a)$$

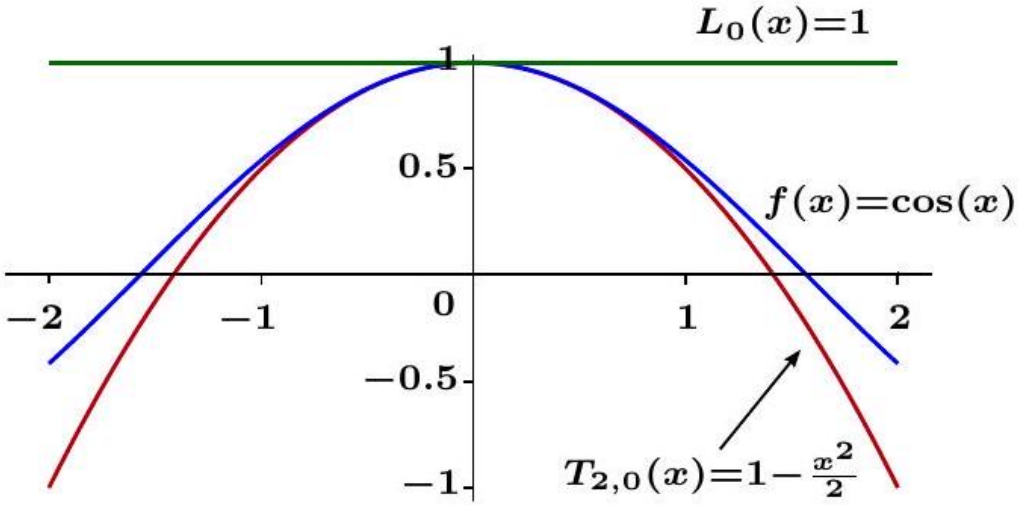
iii) $p''(x) = 2 \cdot c_2 \Rightarrow f''(a) = p''(a) = 2 \cdot c_2 \Rightarrow c_2 = \frac{f''(a)}{2}$

Note: This polynomial, denoted by

$$T_{2,a}(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 = L_a^f(x) + \frac{f''(a)}{2}(x-a)^2$$

is called the second degree Taylor polynomial of $f(x)$ centered at $x = a$.

Taylor Polynomials



Example:

Let $f(x) = \cos(x)$.

Then,

$$f(0) = \cos(0) = 1$$
$$f'(0) = -\sin(0) = 0$$
$$f''(0) = -\cos(0) = -1$$

So

$$L_0(x) = f(0) + f'(0)(x-0) = 1 + 0(x-0) = 1$$

for all x while

$$\begin{aligned} T_{2,0}(x) &= f(0) + f'(0)(x-0) + \frac{f''(0)}{2}(x-0)^2 \\ &= 1 + 0(x-0) + \frac{-1}{2}(x-0)^2 \\ &= 1 - \frac{x^2}{2}. \end{aligned}$$

Taylor Polynomials

Question 3: Can we encode more derivatives?
Answer: If
$p(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3$,
then

1. $p(a) = f(a)$,
2. $p'(a) = f'(a)$,
3. $p''(a) = f''(a)$, and
4. $p'''(a) = f'''(a)$.

Notation: In this case, we call $p(x)$ the third degree Taylor polynomial centered at $x = a$ and denote it by $T_{3,a}(x)$.

Definition 2.2.1 (Taylor Polynomials)

Assume that $f(x)$ is n-times differentiable at $x = a$. The n-th degree Taylor polynomial for $f(x)$ centered at $x = a$ is the polynomial

$$\begin{aligned} T_{n,a}(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \\ &\quad \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n \end{aligned}$$

Observation: Using the convention where $0! = 1! = 1$ and $(x-a)^0 = 1$, we have the following:

$$T_{0,a}(x) = \frac{f(a)}{0!}(x-a)^0 = f(a)$$

$$T_{1,a}(x) = \frac{f(a)}{0!}(x-a)^0 + \frac{f'(a)}{1!}(x-a)^1 = L_a^f(x)$$

$$T_{2,a}(x) = \frac{f(a)}{0!}(x-a)^0 + \frac{f'(a)}{1!}(x-a)^1 + \frac{f''(a)}{2!}(x-a)^2$$

Taylor Polynomials

Key Observation: A remarkable property about $T_{n,a}(x)$ is that for any k between $\mathbf{0}$ and n,

$$T_{n,a}^{(k)}(a) = f^{(k)}(a)$$

That is, $\boldsymbol{T}_{n,a}(x)$ encodes not only the value of $f(x)$ at $x = a$ but all of its first n derivatives as well. Moreover, this is the only polynomial of degree n or less that does so.

Taylor Polynomials for $\cos(x)$

Example: Find all of the Taylor polynomials up to degree 5 for the function $f(x) = \cos(x)$ with center $x = 0$.

Solution: We know that

$$\begin{aligned} f(0) &= \cos(0) = 1 \\ f'(0) &= -\sin(0) = 0, \text{ and} \\ f''(0) &= -\cos(0) = -1 \end{aligned}$$

It follows that

$$\begin{aligned} T_{0,0}(x) &= 1 \\ T_{1,0}(x) &= L_0(x) = 1 + 0(x-0) = 1, \text{ and} \\ T_{2,0}(x) &= 1 + 0(x-0) + \frac{-1}{2!}(x-0)^2 = 1 - \frac{x^2}{2} \end{aligned}$$

for all x.

Note:

$$T_{0,0}(x) = 1 = T_{1,0}(x)$$

Taylor Polynomials for $\cos(x)$ cont'd

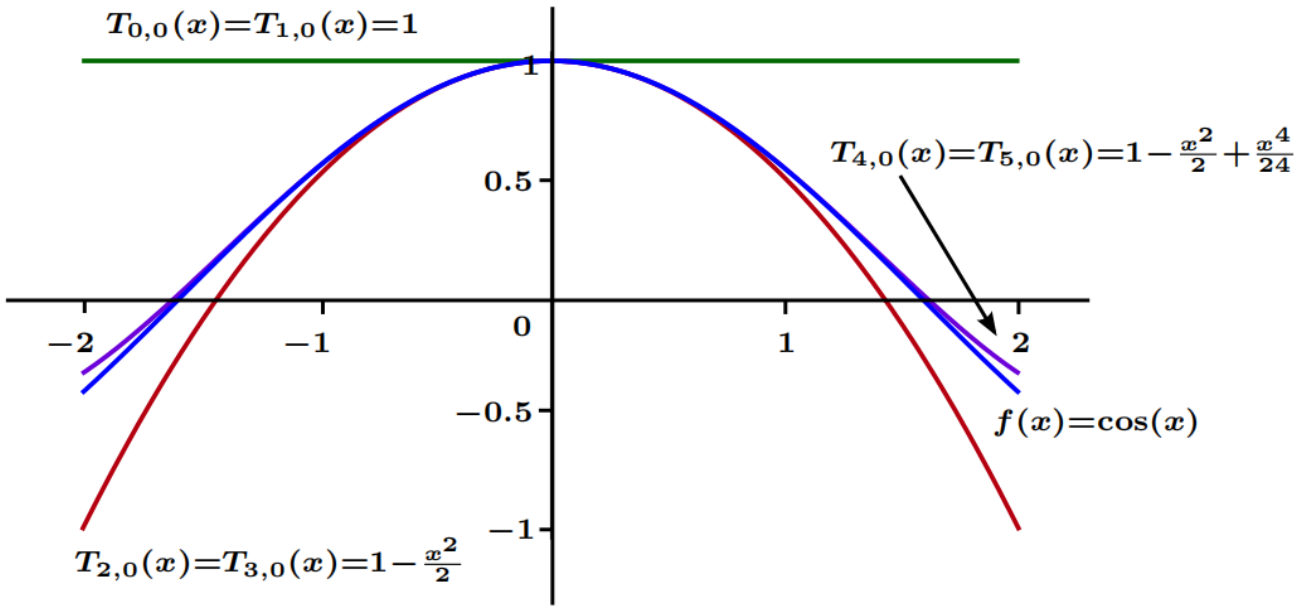
Solution (continued): Recall $f'''(x) = \sin(x)$, $f^{(4)}(x) = \cos(x)$, and $f^{(5)}(x) = -\sin(x)$, we get $f'''(0) = \sin(0) = 0$, $f^{(4)}(0) = \cos(0) = 1$ and $f^{(5)}(0) = -\sin(0) = 0$. Hence,

$$\begin{aligned} T_{3,0}(x) &= 1 + 0(x-0) + \frac{-1}{2!}(x-0)^2 + \frac{0}{3!}(x-0)^3 \\ &= 1 - \frac{x^2}{2} \\ &= T_{2,0}(x) \end{aligned}$$

We also have that

$$\begin{aligned} T_{4,0}(x) &= 1 + 0(x-0) + \frac{-1}{2!}(x-0)^2 + \frac{0}{3!}(x-0)^3 + \frac{1}{4!}(x-0)^4 \\ &= 1 - \frac{x^2}{2} + \frac{x^4}{24} \\ &= T_{5,0}(x) \end{aligned}$$

Taylor Polynomials for $\cos(x)$ cont'd



Note: The diagram displays the graph of $\cos(x)$ with its Taylor Polynomials up to degree 5.
For $k \geq 0$

$$T_{2k,0}(x) = T_{2k+1,0}(x) = 1 - \frac{x^2}{2} + \frac{x^4}{4!} + \cdots + (-1)^k \frac{x^{2k}}{2k!}$$

Taylor Polynomials for e^x

Example: Let $f(x) = e^x$. Then

$$f^{(k)}(x) = e^x \Rightarrow f^{(k)}(0) = e^0 = 1$$

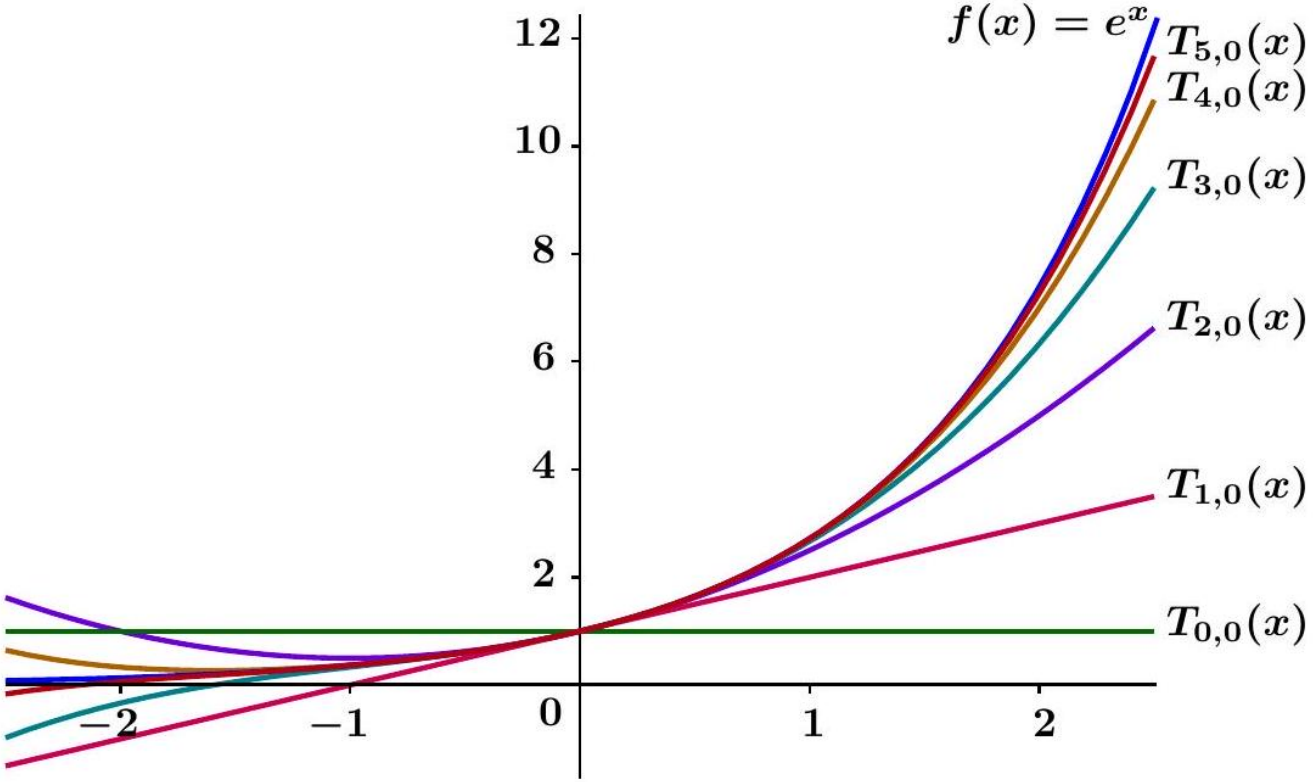
for any k. Therefore

$$T_{n,0}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}(x-0)^k \quad = \sum_{k=0}^{n} \frac{e^0}{k!}x^k = \sum_{k=0}^{n} \frac{x^k}{k!}$$

In particular,

$$T_{0,0}(x) = 1$$
$$T_{1,0}(x) = 1 + x$$
$$T_{2,0}(x) = 1 + x + \frac{x^2}{2}$$
$$T_{3,0}(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$
$$T_{4,0}(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$$
$$T_{5,0}(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120}$$

Taylor Polynomials for e^x cont'd



$$T_{n,0}(x) = \sum_{k=0}^{n} \frac{x^{(k)}}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots + \frac{x^n}{n!}$$

Taylor Polynomials

Recall: Taylor Polynomials

Assume that $f(x)$ is n-times differentiable at $x = a$. The n-th degree Taylor polynomial for $f(x)$ centered at $x = a$ is the polynomial

$$\begin{aligned} T_{n,a}(x) =& \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \\ =& f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \\ & \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n \end{aligned}$$

Observation: We have seen that for functions $f(x)$ such as $\cos(x)$, $\sin(x)$ and e^x, that

$$T_{n,a}(x) \cong f(x)$$

near $x = a$.

Taylor's Remainder

Definition 2.2.2 (Taylor Remainder)

Assume that $f(x)$ is n times differentiable at $x = a$. Let

$$R_{n,a}(x) = f(x) - T_{n,a}(x)$$

$R_{n,a}(x)$ is called the n-th degree Taylor remainder function centered at $x = a$.

Note: The error in using a Taylor Polynomial to approximate $f(x)$ is given by

$$\text{Error} = |R_{n,a}(x)|.$$

Central Problem: Given a function $f(x)$ and a point $x = a$, how do we estimate the size of the Taylor Remainder $R_{n,a}(x)$?

Taylor's Theorem

Theorem 2.2.3 (Taylor's Theorem)

Assume that $f(x)$ is $n+1$-times differentiable on an interval I containing $x = a$. Let $x \in I$. Then there exists a point c between x and a such that

$$f(x) - T_{n,a}(x) = R_{n,a}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Taylor's Theorem

Remarks:

1) Since $T_{0,a}(x) = f(a)$, when $n = 0$

$$f(x) - T_{0,a}(x) = f(x) - f(a) = f'(c)(x - a)$$

which is the Mean Value Theorem.

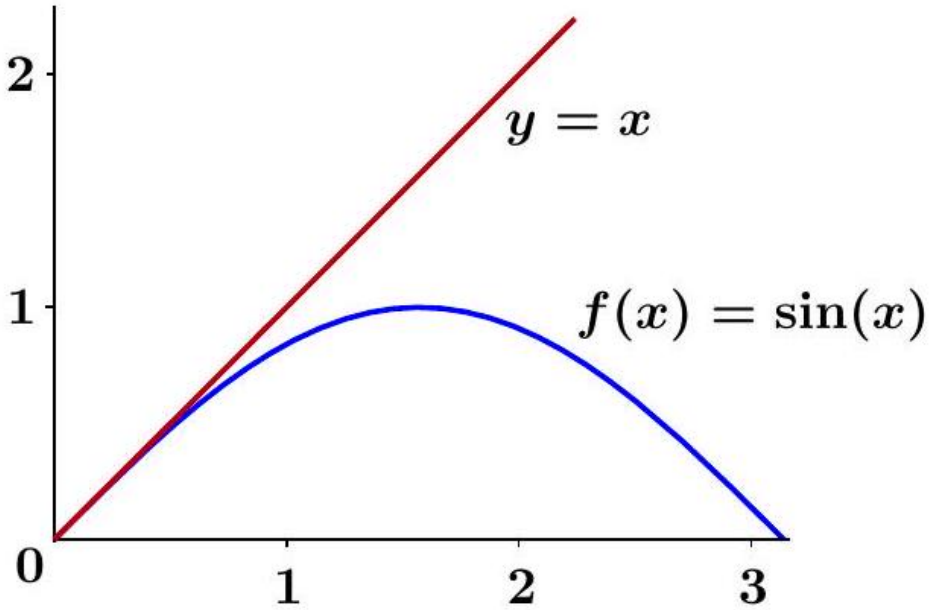
2) Since $T_{1,a}(x) = L_a^f(x)$,

$$\left|f(x) - L_a^f(x)\right| = |R_{1,a}(x)| = \frac{|f''(c)|}{2}(x - a)^2$$

is the error in using the linear approximation.

3)Taylor's Theorem does not tell us how to find the point c, only that it exists. Therefore, to estimate $R_{n,a}(x)$ we must first estimate how large $\left|f^{(n+1)}(c)\right|$ could be without knowing the value of c.

Example



Remark: It can be shown that

$$\sin(x) \leq x$$

for all $x \geq 0$.

We can use Taylor's Theorem to show this for all $x \in \left[0, \frac{\pi}{2}\right]$.

The statement is clearly true for $x = 0$. Let $x \in \left(0, \frac{\pi}{2}\right]$. Then by Taylor's Theorem there is a $c \in (0, x)$ with

$$\sin(x) - x = R_{1,0}(x) = \frac{-\sin(c)}{2}x^2 < 0$$

since $\sin(c) > 0$ for any $c \in \left(0, \frac{\pi}{2}\right)$.

Important Remark: Suppose that $f^{(n+1)}(x)$ is a continuous function on $[a, b]$. By Extreme Value Theorem there are $m, M \in \mathbb{R}$ such that

$$m \leq f^{(n+1)}(x) \leq M$$

for all $x \in [a, b]$. By Taylor's Theorem there is a c between x and a such that

$$R_{n,a}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Therefore,

$$f(x) - T_{n,a}(x) = R_{n,a}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{k+1} \leq \frac{M}{(n+1)!}(x-a)^{n+1}$$

for all $x \in [a, b]$ since c is also in $[a, b]$. Similarly, we can get the lower bound

$$\frac{m}{(n+1)!}(x-a)^{n+1} \leq f(x) - T_{k,0}(x), \quad x \in [a, b].$$

Taylor's Inequality

Theorem 2.2.4 (Taylor's Inequality)

Suppose that f is an $(n+1)$-times continuously differentiable function on $[a,b]$, and denote by m and M the minimum and maximum, respectively, of $f^{(n+1)}$ over $[a,b]$; that is,

$$m \leq f^{(n+1)}(x) \leq M, \quad x \text{ in } [a,b].$$

Then Taylor's inequality

$$\begin{aligned} & f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{m}{(n+1)!}(x-a)^{n+1} \\ & \leq f(x) \\ & \leq f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{M}{(n+1)!}(x-a)^{n+1} \end{aligned}$$

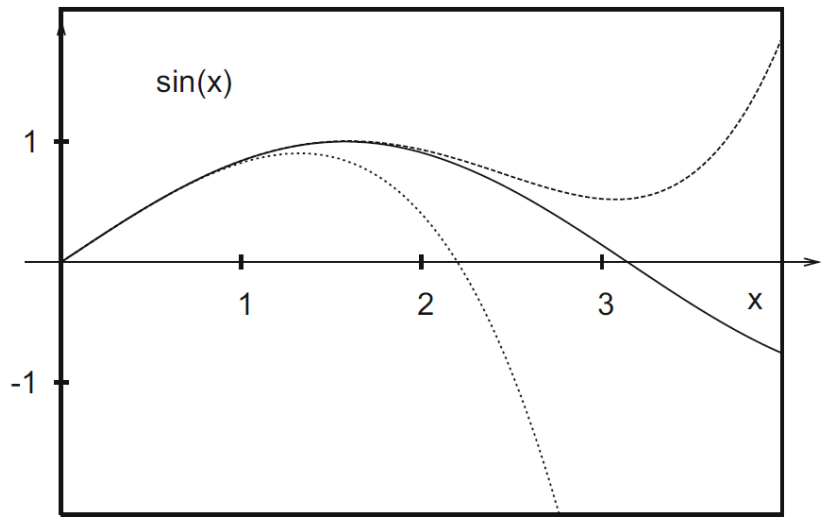
holds for all x in $[a,b]$.

Example

We apply Taylor's inequality for $f(x) = \sin x$ on $[0, 4]$, where $a = 0$ and $n = 5$. The first four derivatives are

$$f'(x) = \cos x, \quad f''(x) = -\sin x,$$

$$f'''(x) = -\cos x, \quad f''''(x) = \sin x.$$



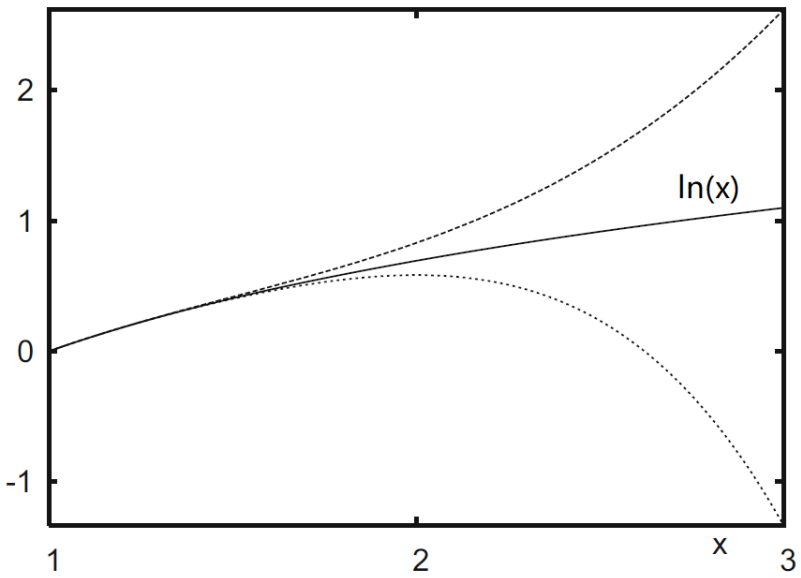
Evaluate at $a = 0 : f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = -1, f''''(0) = 0$. Then the fourth Taylor polynomial $T_{4,0}(x) = x - \frac{x^3}{3!}$. We have $f^{(5)}(x) = \cos x$, so $-1 \leq f^{(5)}(x) \leq 1$ on $[0, 4]$. Therefore,

$$x - \frac{x^3}{3!} - \frac{x^5}{5!} \leq \sin x \leq x - \frac{x^3}{3!} + \frac{x^5}{5!}.$$

Example

Consider $f(x) = \ln x,\ x \in [1, 3]$,
and choose $a = 1$, $n = 4$:

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2},$$
$$f'''(x) = 2!\frac{1}{x^3}, \quad f''''(x) = -3!\frac{1}{x^4}$$
$$f(1) = 0, \quad f'(1) = 1,$$
$$f''(1) = -1, \quad f'''(1) = 2.$$



Since $f^{(4)}(x)$ is increasing,

$$-3! \le f^{(4)}(x) \le -3!\frac{1}{3^4}.$$

By Taylor's inequality, for $1 \le x \le 3$,

$$(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4$$
$$\le \ln x \le (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{3^4 4}(x-1)^4.$$

Section 3

Taylor Series

Taylor Series

Remark: Given a function f that can be represented by a power series

$$f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$$

centered at $x = a$ with $R > 0$, we have seen that f has derivatives of all orders at $x = a$ and that

$$a_n = \frac{f^{(n)}(a)}{n!}$$

So, in fact,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Question: If we assume that a function f has derivatives of all orders at $a \in \mathbb{R}$, then the above series can certainly be constructed. Does the series represent f?

Taylor Series

Definition 2.3.1 (Taylor Series)

Assume that f has derivatives of all orders at $a \in \mathbb{R}$. The formal series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is called the Taylor series for f centered at $x = a$.
We write

$$f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

In the special case where $a = 0$, the series is often referred to as the Maclaurin series for f.

Key Fact: If f is represented by any power series it must be its Taylor series.

Taylor Series

Central Problems: Suppose that f is any function for which $f^{(n)}(a)$ exists for each n.
1) For which values of x does the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

converge?
2) If the series above converges at x_0, is it true that

$$f(x_0) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x_0-a)^n ?$$

That is, can f be reconstructed from the data contained in its various derivatives at the single point $x = a$?

Taylor Series

Example: Let $f(x) = e^x$. Then

$$f^{(k)}(x) = e^x \quad \text{and} \quad f^{(k)}(0) = e^0 = 1$$

for all $k = 0, 1, 2, 3, \ldots$
Hence

$$e^x \sim \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

and this series converges for all $x \in \mathbb{R}$.
Fact: We know that

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

for all $x \in \mathbb{R}$.

Taylor Series

Example: Let

$$g(x) = \begin{cases} \frac{1}{e} & \text{if } x < -1 \\ e^x & \text{if } -1 \leq x \leq 1 \\ e & \text{if } x > 1 \end{cases}$$

Then

$$g^{(k)}(0) = e^0 = 1$$

for all $k = 0, 1, 2, 3, \ldots$ so that

$$g(x) \sim \sum_{n=0}^{\infty} \frac{g^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

and this series converges for all $x \in \mathbb{R}$.
However

$$g(x) \neq \sum_{n=0}^{\infty} \frac{g^{(n)}(0)}{n!} x^n$$

if $x = 7$.

Taylor Series

Important Observations:

Suppose that f is any function for which $f^{(n)}(a)$ exists for each n.

1. Many different functions can have the same Taylor series centered at $x = a$ as does f.
2. Even if the Taylor series of f converges at a point x_0, it need not converge to $f(x_0)$.
3. The Taylor series centered at $x = a$ will always converge at a with value $f(a)$.

Taylor Series

Key Observation: Suppose that f is a function for which $f^{(n)}(a)$ exists for each n and hence with Taylor series

$$f(x) \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Then the k-th partial sum of the Taylor Series is

$$T_{k,a}(x) = \sum_{n=0}^{k} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

which is the k-th degree Taylor polynomial for f centered at $x = a$.
Hence

$$f\left(x_0\right) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}\left(x_0-a\right)^n = \lim_{k\to\infty} T_{k,a}\left(x_0\right)$$

if and only if

$$\lim_{k\to\infty} R_{k,a}\left(x_0\right) = 0$$

Convergence of Taylor Series

Theorem 2.3.2 (Convergence Theorem for Taylor Series)

Assume that f has derivatives of all orders on an interval I containing $x = a$. Assume also that there exists an M such that

$$\left| f^{(k)}(x) \right| \leq M$$

for all k and for all $x \in I$. Then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

for all $x \in I$.

Proof: We know that the Taylor series converges at $x = a$ and that

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(a-a)^n = f(a) + 0 + 0 + 0 + \cdots = f(a)$$

Proof (continued):

Choose $x_0 \in I$ with $x_0 \neq a$. Let $k \in \mathbb{N} \cup \{0\}$. Then Taylor's Theorem tells us that there exists a c_k between a and x_0 so that

$$|f(x_0) - T_{k,a}(x_0)| = \frac{\left|f^{(k+1)}(c_k)\right|}{(k+1)!} |x_0 - a|^{k+1}$$

But since

$$\left|f^{(k+1)}(c_k)\right| \leq M$$

we have that

$$0 \leq |f(x_0) - T_{k,a}(x_0)| \leq M \cdot \frac{|x_0 - a|^{k+1}}{(k+1)!}$$

Since

$$\lim_{k\to\infty} M \cdot \frac{|x_0 - a|^{k+1}}{(k+1)!} = 0$$

the Squeeze Theorem shows that

$$\lim_{k\to\infty} |f(x_0) - T_{k,a}(x_0)| = 0$$

and hence that

$$f(x_0) = \lim_{k\to\infty} T_{k,a}(x_0) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x_0 - a)^n$$

Taylor Series for $\cos(x)$

Example: Find the Taylor series centered at $x = 0$ for $f(x) = \cos(x)$.

$$\begin{aligned}
f'(x) &= -\sin(x) &\Longrightarrow\quad f'(0) &= -\sin(0) = 0\\
f''(x) &= -\cos(x) &\Longrightarrow\quad f''(0) &= -\cos(0) = -1\\
f'''(x) &= \sin(x) &\Longrightarrow\quad f'''(0) &= \sin(0) = 0\\
f^{(4)}(x) &= \cos(x) &\Longrightarrow\quad f^{(4)}(0) &= \cos(0) = 1\\
f^{(5)}(x) &= -\sin(x) &\Longrightarrow\quad f^{(5)}(0) &= -\sin(0) = 0\\
f^{(6)}(x) &= -\cos(x) &\Longrightarrow\quad f^{(6)}(0) &= -\cos(0) = -1\\
f^{(7)}(x) &= \sin(x) &\Longrightarrow\quad f^{(7)}(0) &= \sin(0) = 0\\
f^{(8)}(x) &= \cos(x) &\Longrightarrow\quad f^{(8)}(0) &= \cos(0) = 1
\end{aligned}$$

In fact,

$$\begin{aligned}
f^{(4k)}(x) &= \cos(x) &\Longrightarrow\quad f^{(4k)}(0) &= \cos(0) = 1\\
f^{(4k+1)}(x) &= -\sin(x) &\Longrightarrow\quad f^{(4k+1)}(0) &= -\sin(0) = 0\\
f^{(4k+2)}(x) &= -\cos(x) &\Longrightarrow\quad f^{(4k+2)}(0) &= -\cos(0) = -1\\
f^{(4k+3)}(x) &= \sin(x) &\Longrightarrow\quad f^{(4k+3)}(0) &= \sin(0) = 0
\end{aligned}$$

Taylor Series for $\cos(x)$

Example (continued):
Find the Taylor series centered at $x = 0$ for $f(x) = \cos(x)$.
Hence

$$\begin{aligned}
\cos(x) &\sim \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \\
&= 1 + \frac{0x}{1!} + \frac{-1x^2}{2!} + \frac{0x^3}{3!} + \frac{1x^4}{4!} + \cdots \\
&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\
&= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}
\end{aligned}$$

Note: The series converges for all $x \in \mathbb{R}$.

Taylor Series for $\cos(x)$

Example: (continued) If $f(x) = \cos(x)$, then

$$\left| f^{(k)}(x) \right| \leq 1$$

for all $x \in \mathbb{R}$ and $k = 0, 1, 2, \ldots$

It follows from Theorem 2.3.2 that

$$\cos(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

for all $x \in \mathbb{R}$.

Taylor Series for $\sin(x)$

Example: Find the Taylor series centered at $x = 0$ for $g(x) = \sin(x)$.

We have

$$\begin{aligned}
g^{(4k)}(x) &= \sin(x) &\Longrightarrow\quad g^{(4k)}(0) &= \sin(0) = 0\\
g^{(4k+1)}(x) &= \cos(x) &\Longrightarrow\quad g^{(4k+1)}(0) &= \cos(0) = 1\\
g^{(4k+2)}(x) &= -\sin(x) &\Longrightarrow\quad g^{(4k+2)}(0) &= -\sin(0) = 0\\
g^{(4k+3)}(x) &= -\cos(x) &\Longrightarrow\quad g^{(4k+3)}(0) &= -\cos(0) = -1
\end{aligned}$$

Hence

$$\sin(x) \sim x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

Note that

$$\left|g^{(k)}(x)\right| \leq 1, \quad \text{for all } x \in \mathbb{R} \text{ and } k = 0, 1, 2, \ldots$$

By Theorem 2.3.2, we get

$$\sin(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad x \in \mathbb{R}.$$

Convergence of Taylor Series

Example: Let $f(x) = e^x$ and $I = [-M, M], M > 0$.
If $x \in [-M, M]$, then

$$\left| f^{(k)}(x) \right| = e^x \leq e^M$$

Hence

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

for all $x \in [-M, M]$.
Since the above is true for all $M > 0$,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

for all $x \in \mathbb{R}$.